

Laurent Colbois

POSTDOCTORAL RESEARCHER · VISION-LANGUAGE MODELS FOR FORENSIC BIOMETRICS

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🎓 Laurent Colbois



“To create a positive impact in local communities while following and supporting sustainable practices.”

Summary

Postdoctoral Researcher in Machine Learning and Biometrics with a PhD from EPFL/Idiap. Specialized in computer vision, image forensics, and vision-language models, with a strong focus on reproducible research workflows. Experienced in supervising academic projects and communicating technical concepts to diverse audiences. Seeking to leverage expertise in AI and data science to solve real-world problems in industry.

Work Experience

Idiap Research Institute

POSTDOCTORAL RESEARCHER - VISION-LANGUAGE MODELS FOR BIOMETRICS

- Conducting research on Vision-Language Models applied to forensic biometrics.
- Focusing on reproducible research workflows and rigorous evaluation methodologies.

Martigny, Switzerland

Apr. 2025 - Present

Idiap Research Institute

PHD STUDENT - DEEPPAKES, BIOMETRICS, FORENSIC SCIENCE

- Developed systems for detection of AI-generated images, with a focus on fake passport photos
- Evaluated the vulnerability of face recognition systems to deepfakes
- Presented outreach talks on AI and deepfakes to the forensic science community and to police forces
- Supervised four master theses on score calibration for forensic face recognition
- Published several research code packages targeting experiments reproducibility

Martigny, Switzerland

Jan. 2021 - Mar. 2025

Idiap Research Institute

RESEARCH INTERN

- Created and published a synthetic face recognition dataset using generative AI (GANs)
- Evaluated the applicability of this dataset for benchmarking face recognition systems

Martigny, Switzerland

Jul. - Dec. 2020

Logitech

RESEARCH INTERN

- Developed an automated audio scene classification model for auto-equalization of a loudspeaker
- Implemented a real-time demo running on a microcontroller

Lausanne, Switzerland

Feb. - Aug. 2019

Education

University of Lausanne / Idiap Research Institute

PHD IN FORENSIC SCIENCE (MACHINE LEARNING & BIOMETRICS)

- Thesis: Face morphing attacks in the era of deepfakes: risks, detection & source attribution

Lausanne, Switzerland

2021 - 2025

EPFL

M.Sc. IN COMPUTATIONAL SCIENCE & ENGINEERING

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing

Lausanne, Switzerland

2017 - 2020

EPFL

B.Sc. IN PHYSICS

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing

Lausanne, Switzerland

2013 - 2016

1 Publications

2024	Conference Paper , Evaluating the effectiveness of attack-agnostic features for morphing attack detection	<i>IJCB</i>
2023	Conference Paper , Approximating Optimal Morphing Attacks using Template Inversion	<i>IJCB</i>
2022	Conference Paper , On the detection of morphing attacks generated by GANs	<i>BIOSIG</i>
2022	Conference Paper , Are GAN-based morphs threatening face recognition?	<i>ICASSP</i>
2021	Conference Paper , On the use of automatically generated synthetic image datasets for benchmarking face recognition	<i>IJCB</i>

Skills

Domains	Biometrics, Machine learning, Deep learning, Computer vision, Audio processing, Numerical analysis, Reproducible research, Scientific communication
Programming	Python, PyTorch, Scikit-Learn, Pandas, Unix, Git, Conda, Snakemake, C++
Languages	French (Mother tongue), English (Professional proficiency), German (B2)